

ActInf GuestStream 058.1 ~ Working with Gerald Edelman

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foreign hello and welcome everybody it is September 19 2023 we're here at the octave inference Institute in guest stream 58.1 and today the session will be around the experiences of those who have worked with Gerald Edelman Edelman lived from 1929 to 2014 and as hopefully this conversation can explore and unpack a little bit worked in various areas related to the brain body and mind so without further ado I will have our guests introduce themselves give a little bit of a background on how they came to be working with Professor Edelman and then we have several prepared questions and several other questions so thank you all for joining and perhaps Andrew if you'd like to begin all right so uh I'll just start a little bit with my background I did my PhD at University of Minnesota um and then I did my postdoc at Hopkins with Apostles georgopoulos and Vernon mock Castle and then I got a job in Phoenix working in a small Neuroscience Institute or neuroscience department at a hospital and I worked there for six or seven years and my interest was working with uh non-human primates with monkeys and studying how they uh reached for different objects and Drew objects and then looked at the activity in primary motor cortex neural and we carried out uh different kinds of analysis based on the directionality of the neurons in motor cortex built populations and were able to actually look at these neural signals and see how they corresponded in a pretty exact way to the way the animals are moved in time and space in other words trajectories and um after working there six or seven years I was invited to come and look at this new Institute that Gerald Edelman was establishing in San Diego in La Jolla and met a bunch of super interesting people like Julio tanoni and Olaf's Barnes and uh really their deep intellect and curiosity and creativity is convinced me to move to San Diego um and uh so I ended up moving there and so Edelman uh was able to build this gorgeous research facility uh on the Torrey Pines Cliffs basically and uh it was a magnificent architecture and he recruited a number of people besides uh spawns and tanoni and so he divided the group into theorists and experimentalists and I was an experimentalist but we'd have lunch every day together as a

group and uh discuss different topics and and I was really exposed to a whole new way of thinking about Neuroscience in terms of sort of the cognitive aspects of it which I still have difficulty with to tell you personally because I don't even know how you Define cognition and I'm not sure that anybody really can or as Vernon mock Castle would say I don't know the answer to that nor does anybody else maybe Carl would uh take issue with that um but anyway the coolest thing about it was being really exposed to a lot of creative thinking and deep intellectual Endeavors and I've really never regretted the seven or eight years I spent there I think it really helped change the way my career went um from being sort of a motor control strict sort of uh you know very careful scientist to using my imagination a lot more than I would have otherwise so I'll leave it at that awesome Ching Ming um okay yeah I think the uh speak about uh my my uh Journey with Dr Edelman so I was originally uh from Taiwan I graduated with a medical degree in Taiwan University uh in 78 but I was very interested in doing basic science so I applied the PHD program and got into Rockefeller University and that you know started to search for a laboratory and certainly Dr edman you know with his very powerful intellectual capacity building on the in the building you know it was a ninth floor of the Bronx laboratory I was very impressed so I I joined the laboratory um so um since 78 to 83 I was a PhD student that and I was working on the topic called called neural cell adhesion molecule so at that time element already got his Nobel Prize for Immunology but he would be you know usually you know in use usually elegant way but to say I solved the molecular recognition so he said I don't want to be stay in Immunology to collect more Awards he said I will work out the cell recognition so in his mind that the cell wholesale recognition work is through a group of adhesion molecular interaction uh so that that's how he starts to work on the cell adhesion molecule and the duties house actually work out the chicken cell or tissue molecule so when I joined the laboratory my PhD since was work out um the first mammalian cell adhesion molecule and in addition to end candle I supposed to work out on these um a tissue molecular media seeing the new law cell recognition as he said he would say you know I asked him also after all what would be my PhD thesis and he was the same mean you know when I start to write a poem I don't know how it's going to end you know that's the way he would put his language there so he said just figure out a retinal tactile projection you know so that you know I'm hard working students so I would work out on that and as you know these days our so-called our systems biology approach right is this transcriptome single cell are in the sequencing to identify a non-biased way to identify molecules involved at that time the so-called systems biology approach was making monoclonal antibodies so I would be there

taking out the brain you know and the retina and the tech talk take them and produce I produce 18 batch of this of the monoclonal antibody against these these neuronal cell membranes so I certainly have identified a lot of adhesion molecules but um none of them really show a very distinct pattern I mean these days I think we I mean the field using the transcription factors was able to identify some patterns but at less later it's kind of all altogether but um I I remember that actually no I so anyway that's enough I didn't really figure out the retinal sector projection but that's enough for me to to graduate um but um at the next time though I also still remember you know how I actually um it has a great impact on my professional career so I was cutting chicken spinal cord you know cross-section to mapping resale adhesion molecules so I have some feather chicken feather hanging outside there and they stand with this end came with the most beautiful Exquisite pattern you know I mean brain I data work on the feather as my major research model kind of as a rosetta song to understand the language of morphogenesis but today I remember how I started he said you have this feather they form these branches so these branches are actually originally formed cells then they die to become space but before they die they actually light up with and cam right up with ncam with the most Exquisite pattern there uh so you kind of say okay you're going to carve out this region you light up with and cam you light up with the Sonic and then other signal coming to say you die but before they die they like update pattern so the feather pattern is very Exquisite radio symmetry bilateral symmetry bilateral asymmetry so anyone I was very impressed by those you know those findings and um so that you know that I appreciate that he actually encouraged that he he when he sees something he appreciate he he goes he he encouraged it he didn't say oh you're supposed to work on the brand or whatever he actually like it and subsequently he also you know developed up this idea of couple biology which he would acknowledge to say it's sort of based on the very Exquisite pattern we found in the in the feather but of course he developed it further but some other time he would say if this is not in fact I'm going to I am adamant I'm going to get another Nobel Prize so I I like it let him encourage that direction so later when I became independent I sort of say okay you know um a brand is wonderful but it's all like being curved inside I don't figure it out but I'm not that smart but I might be able to figure out feather pattern formation also I have a reasonable successful career I'll figure out you know the able table of the feather but but as I said not just feather itself but use feather as as a as a rosette custom to figure out the opportunities and uh that sort of I think I appreciate that his encourage of that and and also you know as you know the stem cell biology comes along and the feather in a bird you know baby chicken is one way

about chicken male female all looks differently but they are all from the same group of the same cell that's my lab have demonstrate so it is the so-called stem cell Niche would be modulated by sex hormone by age by season by temperature and then help you produce different feather forms to let them to adapt to the environment so so I I think I I enjoy my research career and it was obviously I have his impact and I think he's intellectually brilliant and always encouraged going out after fundamental principles saying that if you're going to work on the question ask the most profounded questions and approach with the most available technology around the technology of the day you know don't waste your time on small questions and and I think I have to take those in so I can say up to here and we can discuss more later thank you awesome Carl all right sir my name is Carl Friston and like my two colleagues I found my stay with Jerry Adelman both foundational and very very formative um I was quite early in my research career when I joined the Neuroscience Institute um having trained as a psychiatrist I got into brain Imaging and at the Inception of brain Imaging people were for the first time discovering principles of functional brain organizations such as the principal of functional segregation and subsequently principles of integration has segregated um specialized parts of the brain Communicator or coordinated or are integrated together and one of the key theoreticians and neuroanatomists that authored this particular interpretation deployment of neuroimaging with Samir Zeki and semi Zeki for those of you don't know he's he's still very active uh one of the architects of the of newer Aesthetics um in his late retirement now but he was friendly with Jerry Egelman and semi was um famed for um understanding the computational anatomy of early visual cortex and visual cortical hierarchies and the um and I repeat the principles of functional specialization and integration um so after a couple of years um at the Inception of the human brain mapping using positron emission tomography and establishing um this um for example the color Center in the in a particular part of the visual cortex I was effectively sent to New York um so my period of um engagement was a um a fellowship between about 1990 and 1992 93s so this slightly predates Andy um and it covered the um translocation of The neurosciences Institute from the Rockefeller University to um La Jolla um and uh the uh and the auspices of the script so I spent your half my fellowship and the Rockefeller and then subsequently watched the building of the Magnificent neuroscientist Institute on North Torrey Pines Road that Andy referred to with the auditorium if I remember designed by imported experts in audition and Music and Sound Engineering from Sweden it was a really an amazing place I I do remember a couple of lunches before I returned to the UK to pursue my career in imaging Neuroscience um under

the auspices of the welcome trust but only a couple because we were sort of in rented accommodation from the scripts on the other side of the road but it was an awe-inspiring bit of architecture both intellectually and um practically in terms of uh the you know the early days of the Neuroscience um so I was sent there basically uh because um some exactly thought it would be good for me to go and work with Jerry Adelman um retrospectively I now appreciate that I was um sent there to replace a general called Reed Montague who had been tasked with on the pure theory side as opposed to the solidation molecule and the wet lab side of uh adelman's research um on the pure theory side um or the point of my arrival and to a certain extent at the point of Reed's um brief uh exposure um to The neurosciences Institute the theory of neural group selection was becoming Consolidated and in particular what drove the selection of neuronal groups neural Ensemble cell assemblies however you want to describe them and given um edelman's portion for incredibly incisive in um and um straightforward but very creative biological thinking they thought well it's obviously some kind of natural selection um and therefore there must be the homologue of adaptive Fitness that is underwriting the scaffolding and the dynamic um uh structuring of connectivity or connections that were definitive Definitive of in the spirit of functional specialization of segregation but also the mediator of functional um of these neural cell neuronal groups so this um this became known generically as value um so Reed was working on value-dependent learning the plasticity and the principles that underwrite which synaptic connections have the right kind of adaptive Fitness in their milia in that context that enabled them to persist and other connections went away and became or contributed to the you know the sparse coupling of neural networks and and then read and we can talk about the other glorious stories about the the departure of Reed um Reid actually ended up um in a rather circuitous way with uh Terry sonoski at the sulk and subsequently then hooked up with people like Peter Diane and Wolfram um Schultz uh who at that time was in Zurich and of course this the your the story that emerged and that was the reward prediction error story um that um actually ironically uh Reed did not but a number of all from um eventually got the brain prize for some decade or so later but the Inception of that story was actually at the Rockefeller um and it was um basically uh a an attempt by I think Reed to formalize the pr the principles or that must be in play to make the theory of neural group um selection under what were important what was an important um Dynamic from Eileen's point of view which is notion of re-entrant Dynamics so very very early on he had identified the the fundamental points of of recurrent connectivity and re-entry as very formative in um the you know the emergence of these structured groups

and the um Associated functional specialization um our just Open brackets just to point out that you know you could um you could claim that certainly his early conversations with with but people are at that time so preempted stuff in computational Neuroscience that we now would take as of for granted not really realizing where it came from but certainly his work with Vernon Mancus I'm sure Andy will be able to speak to this later on on your the importance of re-entry and Dynamics and oscillations in the formation of these groups and sort of cortical columns and the like that was a sort of I think um you know the monographs that they not the monographs but the the the the papers and the Articles they wrote at that time I think were quite Visionary um Stephen grossberg was also hanging around a few years earlier and I think you could probably claim that between the two of them they invented computational Neuroscience certainly Allah you know New York Boston sort of uh you know the East Coast kind of um uh um style of of computational Neuroscience mostly involved with something called the Neuroscience Research Foundation so that they they as I understand it and again Andy may want to correct me here but this was like the um the Forerunner the society for Neuroscience it was a group of people people um and perhaps Max Cowan is as the name that comes to mind people who actually invented words like Neuroscience uh which wasn't a thing certainly when I was applying to to you know to to my undergraduate studies um and trying to create institutions academic institutions that would endorse this notion of Neuroscience and the Neuroscience Research Foundation was like the old guard and then The Young Turks came along and reacted against the um the Old Guard the Neuroscience Research Foundation to form the society for neuroscience and then we all know the story uh since that time it's now one of the biggest Societies in in Academia um so it's just making the point that Edelman and um you know his colleagues and vicarious colleagues uh you know a lot of people Oliver Sacks being one Ernst Mayer being another of his favorite sort of um characters to talk to all of whom would come and visit and we were privileged as to listen in and conversations In This Very formative time where some of the sort of I think conceptual pillars of the way that we understand your Neuroscience uh um uh you know in the 21st century were um were first articulated probably not using quite the same words that that we use nowadays so no one talks about re-entry anymore it's all about recurrent neural networks or sort of um uh feed forward and feedback and the like but the same underlying Dynamics and maths I think is absolutely there anyway I was brought in to replace Reed Montague um so that was my little contribution um whilst we're in the Rockefeller I made the mistake of um thinking that the right way to do this was to actually write down equations and try and sort of naturalize it in

terms of in terms of mathematics Edelman was a great biological thinker he was not a very good mathematical thinker um so there was a bit of a clash there so I I remember one uh one meeting um very early on when it was being decided what I was going to be doing um and perhaps it's useful just to say the context and you know in which those those early meetings took place so Andy Owen has already mentioned Julio tenoni and her last spawns they were two key uh people uh edelman's young men who already already in inside you um and then I replaced read um and they were working really purely on Theory um and that theory was tested in the context of uh what was uh a forerunner a super Computing a hypercube that was driving in my day little robots so again you know Edelman was doing neurorobotics in the 1990 or 1980s um in exactly the same way that people now in developmental neural neurorobotics are are still addressing exactly the same questions um probably with with the same kind of computer infrastructure that obviously upgraded um so um we were sitting around later amongst uh called out to take a a phone call from from Germany um and there was a look of horror on people's faces because I'd suggested a relationship between value dependent learning and the risk scholar Wagner rule from behavioral psychology um and as soon as I you mention an equation or a formula that set Edelman off and I didn't know that so I didn't know I pushed the wrong button so when the other one came back in I was effectively rusticated I was prevented from doing any work um for three months and for every day afternoon and morning I had to sit in the Rockefeller library and read purely biological great texts and I ended up having to the last one was the growth of biological thought by Ernst Mayer I had to read from from page to page and when I did so I become sufficiently sanitized to be allowed back into the workplace to actually do some work anyway um so that that was our contribution uh and then when we moved to um you know the whole um parade of 16 wheeler tracks is translocating all the kids and all the um all the paperwork to um to La Jolla and then relocating out there settling down um I brought that work to closure and the interesting that was that the point when giulio well Olaf I remember him saying yeah what he wants is basically to understand the and simulate um the uh the brain um in a way that is um draws expiratory power from the architecture of the connectivity it basically would say he wants to be he wants to establish connectomics um as as a discipline and of course you know 10 years later he's Mr connectomix uh Julia at that point was um inventing Integra integrated information Theory um with a particular focus on information theoretic formulations of complexity and what special if you like counter-intuitive behaviors do you get from these biologically configured connectors when you put Dynamics on structure and trying to quantify that in a way that ultimately um

transpired to be measures like Phi in IIT um so that that was a fun time as the math petition the closet mathematician because I wasn't allowed to to say anything about the maths at the time I enjoyed that bit and sort of looking um you know um being in uh at the if you like um Ground Zero at the Inception of all these all these aspirations I mean which eventually led to very fruitful careers and then I went back um to um uh to the UK just had fond memories but and non-fund memories uh but both of a nostalgic Source at some time great Tales chamberger Andrew anything you want to add or I can go to one of the questions well uh just to expound a little bit on what Carl was saying I don't know if you can see this book okay so this was uh something that came out of the Neuroscience research program that was sort of the uh old boys club that Francis Schmidt um started at MIT and later was the basis for the neurosciences Institute and what this was sort of the who's who in Neuroscience so again if you look at this book it says edited by Francis o Schmidt so he was the guy that started that so this persisted uh even to the uh throughout the 2000s um and uh so once or twice a year uh at The neurosciences Institute they would organize one of these meetings and it would be a who's who of Neuroscience so all the Nobel laureates and Neuroscience all the big names would come for uh four or five day meeting and talk about their research and as a member of The neurosciences Institute I got to sit in and meet these people and it was a phenomenal especially for with my background because I was sort of a not on the the mainstream of neuroscience at that point and uh that was a really fascinating so you got to meet people like wolf Schultz and and all the the big names um that were there Francis Crick um you know and uh but what I want to go back to is this original book by Edelman and mock Castle called the mindful brain this is sort of like one of the uh Bibles of early creative thinking about brain function and it free saves the idea that of distributed systems this was sort of the thing that Vernon my castle was very much into um and so the to this day the way most people think about brain function is boxes uh where each part of the brain is confined to a box and then there are arrows pointed to other boxes and so we get this idea that there's a circuitry so you'll often hear neuroscientists talk about circuits and that kind of comes from the idea that you have these boxes and you have arrows and you have plus and minus signs with the arrows and then somehow the pluses and minuses add up and tell you what's going to happen at the next box um and that is nice because you can kind of simplify things and talk about causality in in very concrete terms and it's great if you're an engineer because you can write transfer functions about defining how you can output from an input but that's not the way the brain works and you know I think Kyle can attest to the early days of uh MRI studies where everybody was doing pseudo-colored [Music]

representations of these different areas to see which area would light up in a particular task as if each area had a specific job and could be assigned to a specific task so Neuroscience in general still suffering from that point of view and it's only become a little bit more clear as computational science has matured in the last five or ten years especially with uh you know adaptation of artificial neural networks to the mainstream that you know we can start to explore more biological meaningful ways of brain function and you know Vernon mock Castle recognized that you know it has a sort of the uh I would say the founder of systems Neuroscience modern systems neuroscience and Edelman talked about that a little bit you know there's group Theory and some of the things that Olaf has started um you know with graph Theory still sort of based on strict anatomical connectivity and you know if you read like Olaf's latest review where the field is starting to progress away from that constraint of anatomical connectivity we're starting to get more toward the idea of truly distributed systems and to me that that's super exciting but you can look back to this book you know from the 70s and see that the seed was planted way back then and see how you know almost 50 years later now is finally starting to evolve into something that's becoming more mainstream and if you sort of look at the evolution of Neuroscience I think if you project ahead to the next 10 or 20 years this is going to be really the fundamental change in the way we think about neural function and uh I'm super excited about that and you look and you sort of back in history to see how that's evolved and uh I don't know I I think there's a something very attractive about thinking about things that way um yeah okay yeah okay I'll add something I think talking about the you know Advanced uh Circle so so the ingredients about you know seeing things through I I I don't know in the neurobiology field you people still sort of seeing the new low taoism idea is still there right yeah so that was the time when I was uh with him in Rockefeller University uh he would talk things like that you know uh Darwinism for evolution species and he he saw that in immunological so he applied to the to this new router with some idea uh but I have to say you know subsequently I actually my career is mainly in University Southern California and uh working on this morphogenesis pattern for me tissue pattern formation and in our field in a sort of more bench work you know aspect of element you know this direction not the neural biology Direction but um in the field there were people originally more talking about the stem cells are predetermined and then they go listen yes but we using the skin as a model actually you know I start to write these things kind of a Darwinism within the skin I mean basically is that there is no preset forever stem cells if you wipe out these stem cells other cells are waiting there to take over the space so there is even during development and

during regeneration to make the tissue patterning process robust there is a small so-called Darwinism competition going on in there and the less is sort of also influencing my our work in morphogenesis these days and I think that's kind of the impact of him on this aspect as well yeah wow all right I'll read a question this was from Dave Douglas Dave wrote we know that Professor Edelman use deep analogies in exploring new subjects as in his leveraging his knowledge of immunology and organismal development to probe conceptualization and learning did he ever work specifically on metaphor and analogy as such for example did he leverage his experience of music and poetry to dig out scientific facts while separating fertile from sterile analogies an answer or just what that makes you think of it makes me think of something which Edelman used to call me which I am an intellectual Thug so I'm sure that there it's there's a great answer to that excellent question and it does speak to edelman's skills and interests as a violinist um I do remember that BBC Horizon came to um do a profile on him and his ideas and it was uh prefaced with uh I think shots of of him playing the violin you know to concert standard uh but I have absolutely no idea whether he used those that that sort of part of his life to uh um to um take things Beyond metaphor so as an intellectual Thug um I'm gonna hand over to my two colleagues whether they can give a more elegant answer or or anything in that area of the Arts and uh you brought up poetry anything in the Arts and Sciences well I I have no no biologist if I just say that from my side of the interaction you know I mean he really tried to to merge our science and art and he would think you know a real science would be beautiful and I and that's the way I think when I learned as I mentioned you know when I say oh I I was expecting a more clear-cut CC's idea but he would say you know mean it's like writing a poetry he said that when I start I don't know where it's going to end uh you know it's like some artists when you ask them uh how do you create this painting I I try to ask these questions to some of my friends who are artists and they say they say they do not they do not have a clear blueprint of how that painting what it looks like they have an idea it's kind of floating in the sky and they try to grab it down so they paint on the way and then try to to make it into a painting or I have a friend who are poet I ask them this they say they go up the world to make them come in town and I kind of feel Adam I thought about three to the the scientific idea idea like that you know that's the you know this is my superficial you know interaction yeah uh yeah just some experience so I don't have a correct reply to that but I you know in general um I think there's the idea of creativity that that he brought to the Forefront and then also you know with that creativity there you have to have a certain amount of Courage to move out of the sort of mainstream right so if you have an idea and you have to

have a certain amount of courage to put that forward uh that if that idea is creative and out of the ordinary and I think artists and sort of uh really great scientists have that in common maybe to that Nexus of creativity and career development and courage for you or just in the environment how did his mentoring help bring out these threats in people for better and for worse I I think Carl will could have tested that I I have a hard time viewing Edelman as a as a feel-good mentor I think uh he would throw ideas out there he would be pretty aggressive and uh I would say frightening to young people um so I I wouldn't I want to cast edelman's Behavior as sort of the typical Mentor um but uh again I think the environment that he created was someplace that was a really special and reflected that kind of courageous creativity and I really appreciate that now yeah just to endorse that notion of um I think it's a really nice observation I've often reflected on whether um let's take for example um giulio and Olaf and I think you could probably apply it to me and Andy and a number of other people uh as well but if we just take take Julia and Olaf who both become world leaders thought leaders in their own particular field having brought something new to the table um and in some in some instances in you know more than one area so Julia turnone in terms of um sleep research and synaptic homeostasis um in terms of his working Consciousness studies integrated information Theory um overlap in terms of structural and function and connectomics I mean is the is it that Edelman had a good eye for bright young men or is it that bright young men learned how to do this that this was the way to be courageous and so it's an adversarial in that courage to be creative and to pursue what you have created um in a slightly adversarial or at least sort of defensive way I I I'm not I I don't know but what I do know is um there were certainly a culture that was um established that would allow those kinds of um skills to be developed and you know and is refer to them slightly obliquely and it was uh you know it was a dark culture it was uh and it has to be said it was a very homophilic culture there were no women allowed in this culture so the you know in my time they um the culture was really one man and adoring young men um and you had to um repeatedly affirm that adoration and part of that was um really useful and really enjoyable um and has already mentioned the uh the lunches that exactly the same regime um you know characterized my stay there was a you know a rigorous routine where all the young men would go off to the New York Cafe would have onion rings and and burgers uh and Adam and wax lyrical perhaps this is not art and creativity but it was he was um he was artistic in his joke telling so he'd always have a new joke every every lunch time in a New York uh Diner and that was a routine and it was just pure theater day after day after their incredible energy but it was theater the young men were the audience um and we

basically I think were um dancing to you know his orchestration he was the conductor and he was the director if that culture if that it's possible that that culture did actually instill in all of us and particularly Olaf and giulio um the notion that to make it to make a difference and to be true to one's convictions you really have to sort of fight for and be autonomously pursue what you want to do and it is it's interesting that the the style in which both Julia and Olaf have pursued their their scientific ideology and ideas and creativity is very much is it's slightly Lonesome you know the um with the exception of Julia's work um in um in the microbiology of sleep um the theoretical contributions are very sort of you know associated with them personally and to a certain extent with me and um you know what one could argue is um then you know the um the mathematical version of value-dependent learning which would um be the free energy principle all of these things have the same flavor as neurodynamism or as some people used to call it neural neural edelmanism um and so I I often asked that question was it what was this the skill of Edelman in choosing people who are going to make a difference or did he basically provide a culture in which it gave people the the tools or the um the motivation or the confidence that you can make a difference because you saw you saw him make a difference I don't know um that yeah I think most of you actually you know said something I I share the experience definitely he set up a right environment uh but just like when we do stem cell stem cell and it's niche you need the interaction you need a conversation between episcidia and the messenger cells or in the brain I heard that it's more like interaction between neural cells and we are sales so a successful interaction with the leader whatever the process to move forward and but not every process would be would be would be successful so as as I said I also you know like the way Kyle mentioned about the animals is is the conductor he actually said that in our lab meeting at the time he said I and Adam and I am the conductor you know I you know you guys you know it's my my Symphony prayers and uh and you you you pray together you can make a good music um but it's also true uh not every interaction works because I was with him for almost eight years or so uh not every person worked let's put that away that is a fact uh so some work some don't right so I think people are given that opportunity and uh as Andrew said it's a good outstanding environment so so you pick up the the idea from adamant you develop yourself and some of them you know or what you can tell call it the element level and the medicine then we we success in a certain way or not um but I think it's a it's an exciting environment uh I would put it put it way it's uh it's both you know he inspires you he gives you a sort of a exciting environment to evolve but not everyone is so he can choose everyone to be successful no it's not like that but he did give you that

opportunity to to be successful and to to pick up something and he also give you that courage to do it as I mentioned you know people would normally think you know using the feather as a model is kind of very very high risk but but he's he has that encouragement and I I do that and I I feel I really using fatter as a rosetta song I have learned something for the morphogenesis uh pattern formation field yeah so I appreciate him on that yeah and true or I can ask another question uh I really don't have anything to add right now thank you though it's really amazing it's it's such a different world and the environment of learning and of carrying out science is so different so personally I really appreciate just even a window into settings that are elsewhere and in a different time so thank you for all of this all right I'll ask another question from Dave Professor Edelman was reputed to be skeptical about the risk of abusing mathematical or computer models in creating theories of neural functioning cognition and Consciousness some reviewers went so far as to mock him in print for using computers as research tools but he worked with tanini on the foundations of integrated information Theory an extremely math heavy theory that sometimes explained using Computing imagery did Edelman change his attitude towards the role of math or of computing paradigms and thinking about the mind or did he simply start trusting his colleagues to think computationally without falling into rabbit holes dug by Von Neumann and touring or how did you see the role of math and computers in edelman's thinking and research develop over a period where they were also undergoing such rapid developments more broadly so I I'm pretty sure that Edelman respected mathematics and computational approaches to neuroscience uh he certainly there certainly were a lot of theory fellows that were computationally Savvy uh while I was at the Institute um and uh so for instance uh Carl mentioned these uh robots that were used to sort of explore and learn sort of the beginning I guess of embodied cognition Carl you would say um where they explore the environment and learn and then from that learning they they build internal models so that is uh I would say and that was Olaf's Valley wig I would say that that was highly computational um and certainly a major thrust in edelman's but emphasis and what he wanted to do at The Institute and there were a number of other things I think the integrated part that uh Julio did I think that was almost all Julio's work along with some of the other computational fellows uh so uh you know I think Edelman wanted to be part of that but that really I think was mostly Julio's and uh I I think you know as part of the culture uh Edelman managed to get his name on Julio's paper we used to laugh about that quite a bit um but you know again I I would say that was 99 Julio's work yeah I I can come cat because I was there but when the work was actually done there was a um there was a sort of areaed I can't remember there was some

hypothesis that pertained to a core a dynamical core as um sort of oppressions are necessary for some kind of Consciousness and I think Edelman was comfortable with this sort of intuitive notion of a dynamic core but when you when it became articulated in terms of information theory he um started I he would I think um even if he didn't say so out loud he would have responded as he responded to me with oh you've got mathematosis so hey this is this is another of his sort um ways of dealing with people who had a tendency to over mathematicalize and oversimplify as a physicist toward you know they're reducing a horse to a point and then writing down differential equations he was very allergic to that you know he didn't like that and would call that um an instance of mathematosis I think that's very distinct though from what Andy was just talking about which is the commitment to physically realized embodied computation um and you know I think it's a really well absurd point that you know that this was embodied cognition um Circa 1980 1990. so he was quite Visionary in um I think sort of what testing out ideas in in physical and realized systems you couldn't just hand away with maths you had to show it worked this idea worked in a real um in a real setting and at that time they were um and you know um you know the Darwin series and you know to the extent that um you know I for example had to spend six months learning to paralyze c um in order to program the robot so you know it wasn't he was frightened of new technology or computer science I think he didn't like trivializing things um you know ala physicists with with mathematics but you would certainly um be fully committed to the notion that you that the computation is at the heart of the kind of dynamics that he wanted to understand that and and underwrote theory and indeed I think that you know the theory of neural group selection um speaks that he wanted to see it realized uh you know realized in in a robot so um but yeah I think and is absolutely right at the point when uh Julia sort of just basically um we went straight from the axioms to Consciousness and Phi with equations I I think that was the time that Julia had had to leave the family basically um and and Julie did so brother extrapolated that a little bit more I think you know when I read edelman's books and and some of his papers I don't understand his words okay and you know his phrases and his descriptions you know a lot of things don't make sense if he had written those out or someone had written those out as equations okay what you know that's the nice thing about equations once you see the equation you know you don't have to worry about the word so much uh you know things become pretty clear and I think you know looking back if he'd been able to do that I think his ideas first of all would have been crystallized in his own mind better and second of all would have been able to be others would have understood them more clearly and I think that's really important if you

have for me okay maybe with an engineering background or something you know if you have a model or if you have an idea if you can write that out explicitly with an equation it really clarifies things it crystallize this thing and that was the thing that I liked about Julio's work because again this word of Consciousness you know it's like nobody knows what that really means but at least Julio had a form of equations right where he was talking about information Theory and in a way you know you could have a a cluster where information was internal versus the external information and say that okay if you have to have Consciousness here it is and he could describe that with a concrete equation I mean that's a step forward and again I always say as a someone that that records neurons in the brain all the time if I wanted to find Consciousness how would I know it if I found it right I mean if I had an equation maybe I could do it but you know just the way it's defined right now I mean who knows I could have already discovered it I wouldn't know so uh I I think that's a an interesting point anyway I can't resist just following up on that because um especially evident for me in the remembered present this this notion that his words don't actually make any sense I think that's pure poetry and pure Brilliance because what I tend to find myself doing is writing reading the meaning into his ambiguous sentences and finding all sorts of um meanings which wouldn't have occurred to me otherwise I think that's part of his art I suspect that he would have been terrified if somebody had actually come along and tried to time down to a clear formal naturalized mathematical formalism because he wouldn't have had that sort of biotic expressiveness and ambiguity that of course is at the heart of uh of his evolutionary thinking but I think it's it you know it's it's great that you and you're able to say you don't understand what he wrote Because I I look at some of his sentences like what the Earth is he trying to say here and of course but the exercise of trying to make sense of edelmann itself brings to the table uh some really interesting um sort of associations and ideas I I'm wondering whether that was part of the secret of his success is that he resisted the temptation to be crystal clear and just left that mystical magical ambiguity hanging in the air for people like you and me to to worry about and try to resolve I I suppose he's given that privilege after winning a Nobel Prize I I think uh some of us mere mortals wouldn't be afforded that uh luxury all right I'll read a few more questions we have a few more things so small questions um in the book Topo biology changming did you draw the feathers or is that your work feather is my work yeah it's based on my work yeah well how do we take things forward in science and in our work here in this level of diffractal with this amazing story and history with Edelman but how do we take our historical and our mentoring context like how do you carry that forward is it something

you think about I personally do and I congratulate myself every day on not being like Edelman when it comes to mentoring how so I'm not going to answer the how so question but you could read between the lines whether that's the right thing to do or not I don't know I don't think I don't think you could be Edelman now um you know in 21st century uh with we are um you know we'll be too close to all the culture wars I think to to even entertain that style anymore and I suspect that's quite a good thing although you know pressure makes diamonds um and as Andy says it was an exciting one both my colleagues have said it's an exciting informative time it was a pressure time it was a dark time for me um you know in terms of the system the the Dynamics the systemic the the the personal interactions um but because it was pressured it made diamonds um and so we may be throwing away the opportunity to make academic diamonds and be truly creative in terms of making things that endure like diamonds by not having that style of mentorship but I don't think it would be even legal nowadays to be quite honest to run a lab like a human run so again I mean I I think that over the years Neuroscience yeah I don't know about the rest of science but it's almost become sort of industrialized so there's a method you know you as a young person um you when you finally get your own job you have to set out and if you're at a university environment there's certain boxes you have to check off you have to get a grant you know you have to get certain number of papers you have to get something in science or nature in order to get promoted you know there's there's these steps and I think young people are so focused on moving through those steps at the point in their career where they're going to be the most creative and their minds are the most flexible and they can be the most productive and you know I see that as as a shame actually that that they're not given this opportunity to be a little Freer in the way they accomplish things and then you know when they do accomplish a modicum of success they rely on their past Behavior so instead of saying okay I had to go through that in order to get tenure now that I have tenure I'm free and I can do something that I really want to do that really doesn't happen because people get stuck in a rut and uh I see that as really holding back uh progress in in neuroscience and that was kind of the cool thing about the Neuroscience Institute for better or worse we weren't allowed to go out and get our own grants we had to rely on internal funds but as a result the only person we had to convince was Edelman in order to get funded and uh to a certain extent that allowed us to do things that were really creative in hindsight I think that was a mistake because financially that was a doom for The Institute there could have been a hybrid approach that would have been more productive and but then again uh as Carl was saying the uh the investigators would have had to have more freedom to do that and that was

not exactly in the culture although I think I was an outlier because I maintained a separate lab so I I was somewhat inoculated from that and that made it a lot more pleasant for me um but uh again I see that that you know in current neuroscience it would be nice if people were willing to take more chances um and be a little more creative instead of hoeing to the status quo that would be uh I think uh what we need in order to move forward yeah that um no I I think that in terms of the mentorship hunted you know cow was talking about I mean certainly I I have seen people you know do very well but but I also have the opportunity you know sort of when I moved to Southern California University Southern California to uh have some interaction with the Dr Seymour benzer and uh some of you probably know him I mean you would see actually he is also very successful but but he he when he came out he would always have his students surrounding him in a very very kindly uh I think that you know Adam and um is brilliant but it's kind of the sun uh it's too bright sometimes people have to be distilled into the as you say academic time and let's let's interesting so we also have seen some people you know didn't get along with him uh well but still I think they are and any many impacted here he certainly have encouraged many people go different directions and um in terms of the academic um the the academic future right I mean I think you two are more talking about the neural biology side but on the on the morphogenesis side allow I would also say he always emphasized the patterns you know thinking that you know pattern is kind of the way uh nature you know either biology non-biology they leave some some clue for us to figure out and when I was there I remember every time he talked about the the visual columns uh thousand whistle they have workout he got very excited and in fact I have a one of my peers at that time was working on on the visual Columns of the cat but I think that's favored let's work it never really go out to be published but anyway more practical things I work on the tissue tissue patterns and I I feel that you know go you know you mentioned about the top of biology and I again I think he coined the term and I think the term I mean I think it means a lot because I kind of you know eventually my career was trying to figure out uh these signaling molecules uh because at that time people only talk about the signaling molecule very very important but but the idea of talk about biology is that you even position this simply by positioning this signaling centers in a different topology you can actually have a lot of consequence um so so from there and the signaling Center can also have a different lifetime the strength and the duration and the the the range they can inference so by money my by modulating you know he likes these terms so modulating this signaling Center you can create a lot of regional differences and up to now as I say from my own take you know but you look at it right feather

here Downing feather here you know the beautiful crown feather here tail feather but these are all from the same genome so it actually is through the positioning of these different signaling Center in in a different way and you you basically have the same same ingredient but you can play with them by topologically temporal space temporal and especially arrange them and to to create a quite complex pattern so so I I feel that in the tissue patterning morphogenesis side now I I feel that you know his vision is is great and I I think we still try to work out more about how these patterns are positioned but I think that's sort of the direction in a more visualized purple biology aspective we can appreciate awesome well perhaps if each of you would like to give any closing thoughts yes I'll I'll just do it to Echo you know those last sentiments I think everything that Edelman said or try to say um depending whether you could understand him I think he's absolutely right and we're slowly discovering the truths we just heard about terminology and the importance of patterns and of course if you speak to somebody like Mike Levin he'd be talking about exactly the same thing he'd use words like basal cognition and you know pattern formation but he I think that we're talking about the same uh the same principles of biological self-organization um you know and Adelman I I didn't know about turbo biology um but I do know you know there are other fundamental ideas that he just you know incidental to him I am absolutely sure I think Notions of degeneracy in the neurosciences and psychology um you know the notion of many to one structure function mappings for example these are fundamental ideas which now are taken almost for granted um in in my world but you know you sort of forget how much he actually contributed um or at least laid the seeds for and sometimes very explicitly so it should be acknowledged although he was as Mark Reichel uh put it in his typically gentlemanly fashion Edelman was a very complicated man it was also very brilliant and his responsible I think for a lot of the direction of travel and the intellectual growth certainly of my intellectual growth but I would say also many aspects of of our academic communities thank you Carl perhaps Andrew oh wait unmute and then please continue so I I would say you know uh if there's anyone listening out there there's something to be said about maybe looking back and in ancient literature that goes back quite a bit and maybe seeing how some of these things came about and trying to put it together and see that you know there's this iteration that constantly takes place people ReDiscover the same thing that's been discovered multiple times um and you know try to see how there is uh a status quo in the way science works and then there's these outliers where there's sort of creative bubbles that that come before and I think uh it's interesting to think of Edelman and the things he did and the people that were associated with

him in terms of sort of these disruptive right they call it disruptions um and uh how those disruptions may have consequence many years later it's it's kind of interesting if you take the Long View so I I think it's um maybe worthwhile thinking about this once in a while and for young people out there to be encouraged to think off to think out of the normal way of doing things once in a while try to Foster your creativity and have the courage to to go after some of those ideas I think that's a take-home message for me anyway yeah and I I would say I always uh like you know all sort of you know influenced about his ideas saying that we should study the important question that have a general meaning so I I think that kind of idea she said that look at the big pictures I remember when I was there you talk like that big pictures so that is where when he thinking about the Darwinism he would say it's not only for the species Evolution for the for the immunological selection he was push it to to the brand Theory and and for me I work on the morphogenesis I actually would come to realize that is also what happened at the tissue level these cells are competing with each other to form that patterns and then pattern is selected when there is a functional Advantage it will be civilized so so I think the idea of saying if we're going to do science we do we ask big questions ask for them generalization I I appreciate that and certainly he is like how you want to call it right a huge elephant so we all touch him through different side and have a different take on that and in terms of the young scientists I also remember he likes the Chinese proverb I I cited to him unless he's to say if you wanted to get a Tiger's son uh you must get into Tiger's cave so so so we treat it thank you all that's a great no really appreciate this conversation and again thanks until next time thank you all right bye yeah nice meeting you all I'm David bye-bye thank you